3.1 Classwork: Sets and Venn Diagrams Review — Markscheme

Total 82 marks.

1. [Maximum mark: 13]

   (a) $P = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$ **A1**

   $n(P) = 10$ **A1**

   Common errors: including 1*, or including* 9*,* 15*,* 21*,* 25 *or* 27 *(odd but not prime): A0 for the list; FT for* $n(P)$*.*

   (b) 2 **A1**

   Every other even number has 2 as a factor as well as 1 and itself, so it has more than two factors. **R1**

   (c) (i) false (ii) true (iii) false (iv) true (v) true **A1A1A1A1A1**

   (d) $F = \{1, 2, 3, 5, 6, 10, 15, 30\}$ **A1**

   $F \cap P = \{2, 3, 5\}$ **A1**

   the prime factors of 30 **A1**

   (e) infinite **A1**

   Teaching note: Euclid's proof. If $p_1, \ldots, p_k$ *were all the primes,* $p_1 p_2 \cdots p_k + 1$ *would have a prime factor not in the list.*

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2. [Maximum mark: 12]

   (a) (i) false (ii) true (iii) false (iv) false

   (v) true (vi) true (vii) true (viii) false **A1** *each*

   (ii): in the IB, $\mathbb{N}$ *includes* 0*. (vi):* $0.\overline{3} = \frac{1}{3}$*.*

   (b) **A4**

$\mathbb{N}$:	$7,\ 0,\ \sqrt{16} = 4$
$\mathbb{Z}$ but not $\mathbb{N}$:	-2
$\mathbb{Q}$ but not $\mathbb{Z}$:	$\frac{1}{3},\ -0.6$
$\mathbb{R}$ but not $\mathbb{Q}$:	$\sqrt{5},\ \pi$

   A1 for each correct row. Common error: $\sqrt{16}$ *placed outside* $\mathbb{Q}$ *because it is written with a root sign.*

3. [Maximum mark: 13]

(a) (i) $A = \{-2, -1, 0, 1, 2, 3\}$, $n(A) = 6$ **A1A1**

(ii) $B = \{0, 1, 2, 3, 4\}$, $n(B) = 5$ **A1A1**

Common error: leaving out 0 *(it is in* $\mathbb{N}$*).*

(iii) $C = \{1, 2, 3, 4\}$, $n(C) = 4$ **A1A1**

(iv) D is infinite: there are infinitely many real numbers between 1 and 2, so they cannot be listed. **R1**

(b) (i) open circle at -1, filled circle at 3, the segment between them drawn **A1**

(ii) four separate dots at -2, -1, 0, 1 (no segment) **A1**

(iii) open circle at 0 with an arrow to the left; filled circle at 2 with an arrow to the right **A1**

(c) (i) $\{x \in \mathbb{R} \mid -2 \leq x < 4\}$ **A1**

(ii) $\{x \in \mathbb{Z} \mid -1 \leq x \leq 3\}$ **A1**

(iii) $\{x \mid x > 1\}$ **A1**

Accept $\{x \mid -2 \leq x < 4\}$ *in (i) (*$x \in \mathbb{R}$ *is understood). In (ii) the set must say* $\mathbb{Z}$*; accept* $-2 < x < 4$ *or other equivalent inequalities.*

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4. [Maximum mark: 6]

(a) $n(A) = 9 + 4 = 13$ (b) $n(B') = 9 + 5 = 14$ (c) $n(A \cup B) = 20$

(d) $n(A \cap B') = 9$ (e) $n\big((A \cap B)'\big) = 25 - 4 = 21$ (f) $n(U) = 25$ **A1** *each*

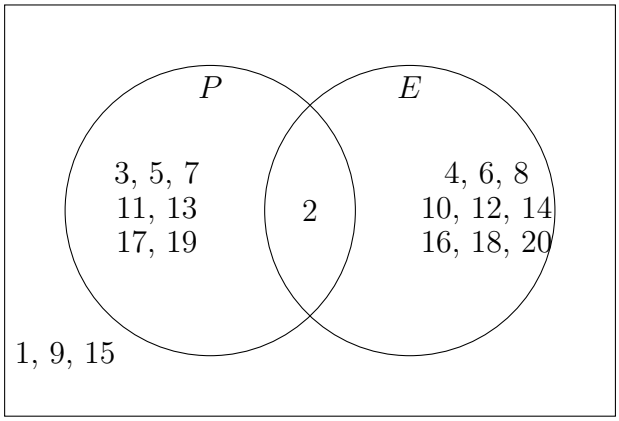
Common error in (b): $n(B') = 9$ *(forgetting the* 5 *outside both circles).*

5. [Maximum mark: 12]

(a) $P = \{2, 3, 5, 7, 11, 13, 17, 19\}$ **A1**

$E = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$ **A1**

(b) **A1A1A1**



A1 for 2 *in the overlap, A1 for* 1, 9, 15 *outside both circles, A1 for the remaining elements correct.*

(c) (i) $P \cap E = \{2\}$ **A1**

(ii) $P \cap E' = \{3, 5, 7, 11, 13, 17, 19\}$ **A1**

(iii) $(P \cup E)' = \{1, 9, 15\}$ **A1**

(d) $n(P \cup E) = 7 + 1 + 9 = 17$ **A1**

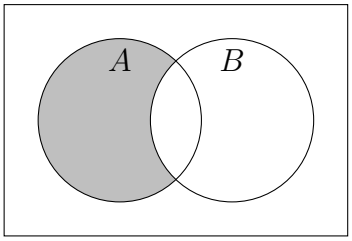
$n(P) + n(E) - n(P \cap E) = 8 + 10 - 1$ **M1**

$= 17$ **A1**

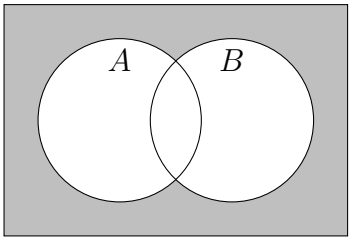
(e) $\frac{3}{20}$ **A1**

FT from their $(P \cup E)'$.

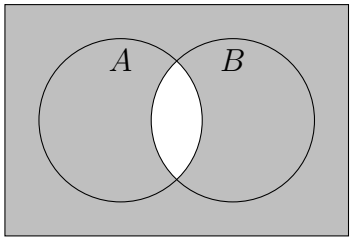
6. [Maximum mark: 6] **A1** *each*



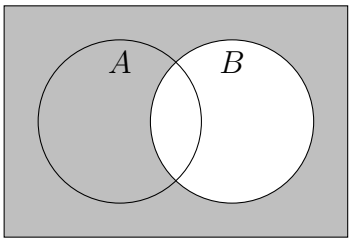
(a) $A \cap B'$



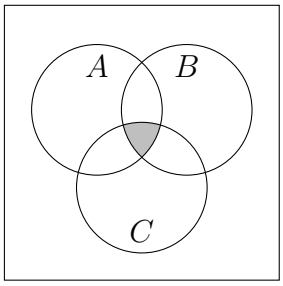
(b) $(A \cup B)'$



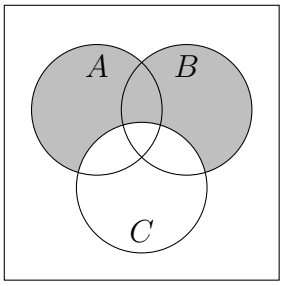
(c) $A' \cup B'$



(d) B'



(e) $A \cap B \cap C$



(f) $(A \cup B) \cap C'$

Common errors: (c) shading only the outside region, which is $(A \cup B)'$; (d) forgetting the region outside both circles.

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7. [Maximum mark: 7]

(a) $A' \cap B$ (b) A' (c) $A \cup B'$ **A1A1A1**

(d) $(A \cap B') \cup (A' \cap B)$ **A1**

Accept $(A \cup B) \cap (A \cap B)'$.

(e) $A \cap B$ (f) $C \cap A' \cap B'$ **A1A1**

(e): the region includes the part of $A \cap B$ inside C, so not $A \cap B \cap C'$. (f): accept $C \cap (A \cup B)'$.

(g) the elements in exactly one of A and B (in A or in B, but not in both) **A1**

8. [Maximum mark: 7]

(a) $s = 6$ **A1**

(b) $19 + 15 - q + 6 = 32$ **M1**

$q = 8$ **A1**

Accept $n(E \cup H) = 32 - 6 = 26$*, so* $q = 19 + 15 - 26$.

(c) $p = 19 - 8 = 11,\ \ r = 15 - 8 = 7$ **A1A1**

FT from (b).

(d) $\dfrac{11 + 7}{32}$ **M1**

$= \dfrac{18}{32} = \dfrac{9}{16}$ **A1**

Common error: $\frac{26}{32}$ *("at least one" in place of "exactly one"): M0A0.*

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9. [Maximum mark: 6]

(a) $65 + 30 - x + 15 = 100$ **M1**

$x = 10\%$ **A1**

Venn diagram: M only 55%, both 10%, D only 20%, neither 15% **A1**

(b) (i) 0.2 **A1**

(ii) $0.55 + 0.2$ **M1**

$= 0.75$ **A1**

*Accept percentages (*20%*,* 75%*) or fractions. FT from (a).*